



Multifocal Deep Learning Model for accurate Cell Segmentation in Traction Force Microscopy

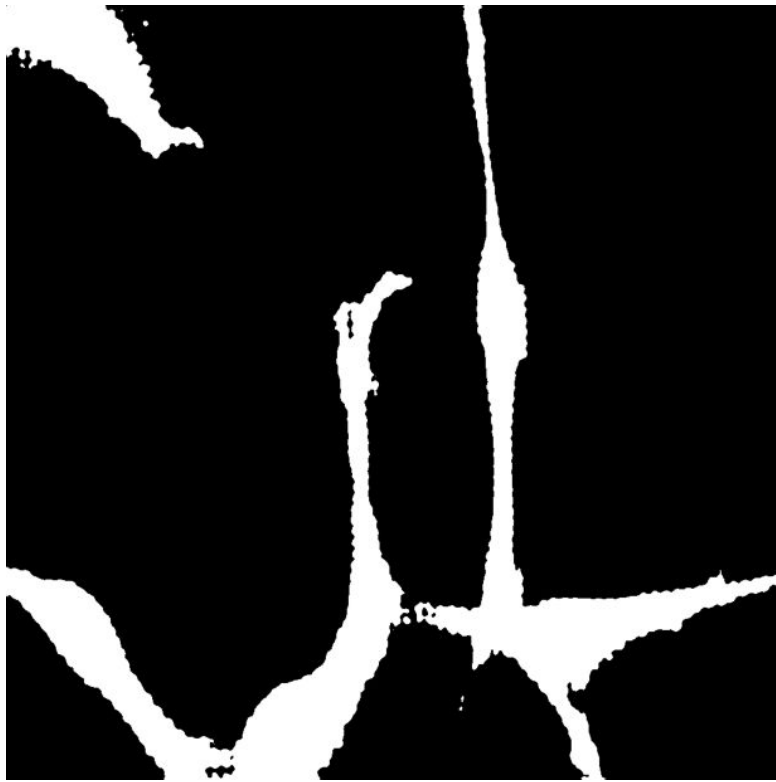
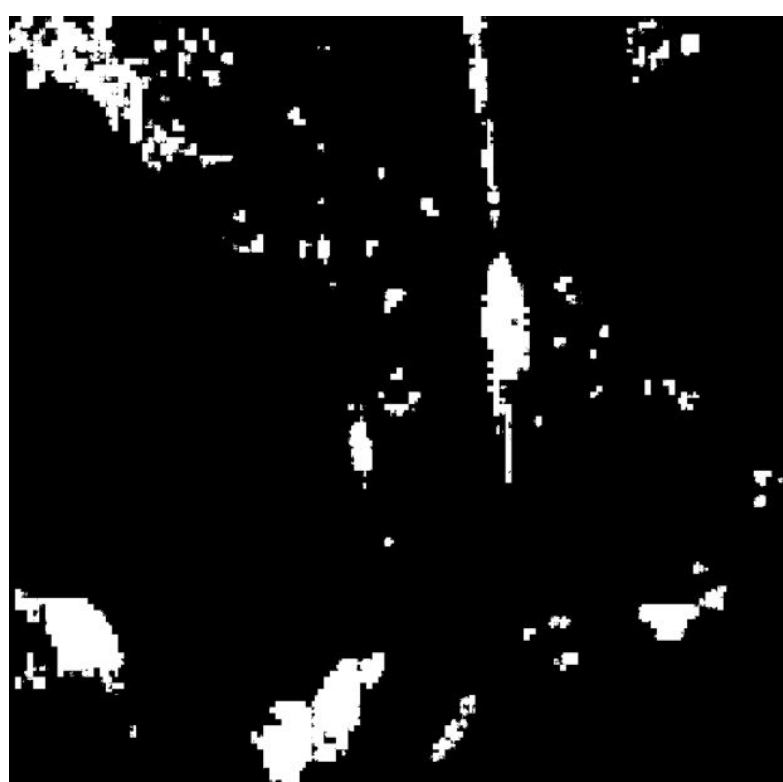
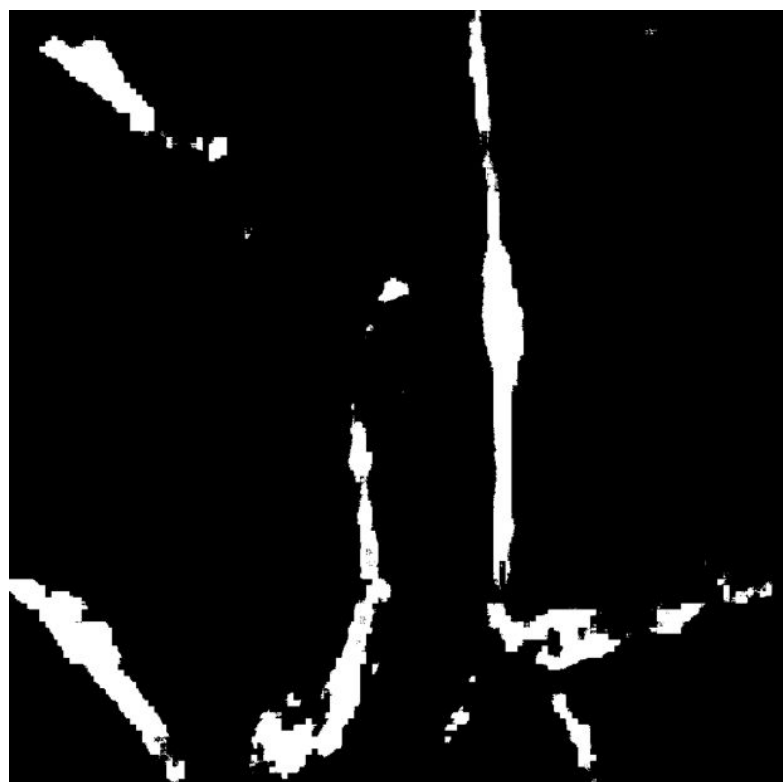
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02456 DEEP LEARNING, DTU COMPUTE, FALL 2024, Technical University of Denmark

Introduction

Our project focus on Multifocal Deep Learning for Accurate Cell Segmentation in Traction Force Microscopy leverages deep learning to achieve precise cell segmentation across multiple focal planes. Traditional segmentation techniques often rely on fluorescence labeling, which can interfere with experiments and may not always be feasible. Our approach aims to eliminate this dependency by directly analyzing brightfield images using deep learning models.

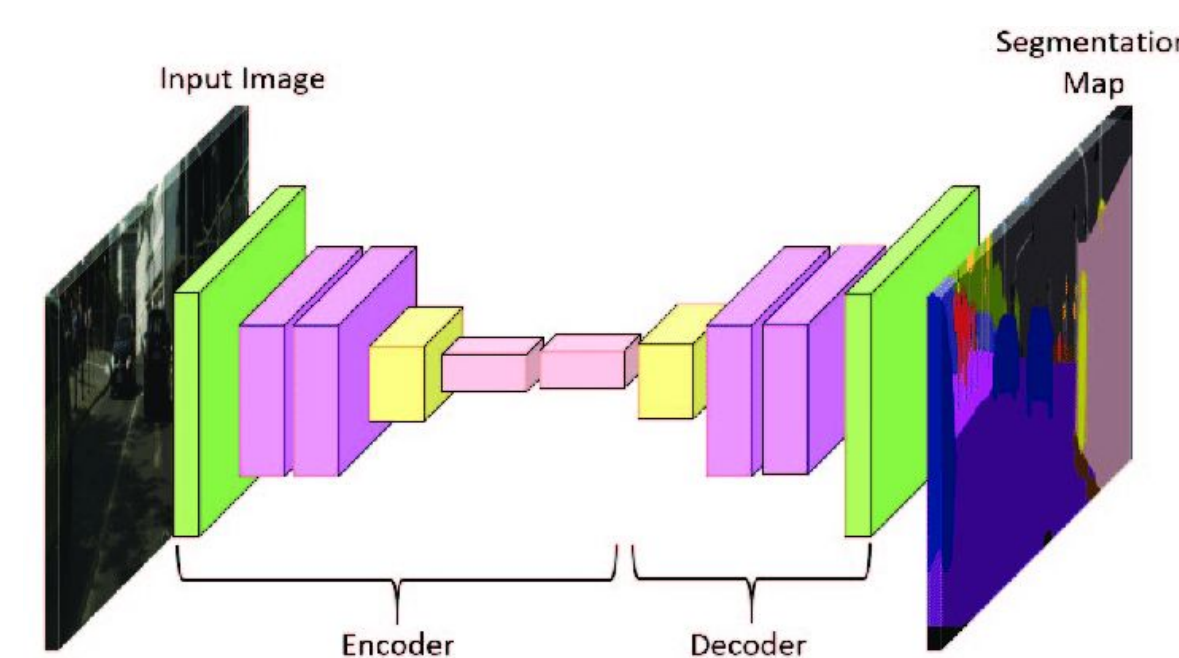


Results

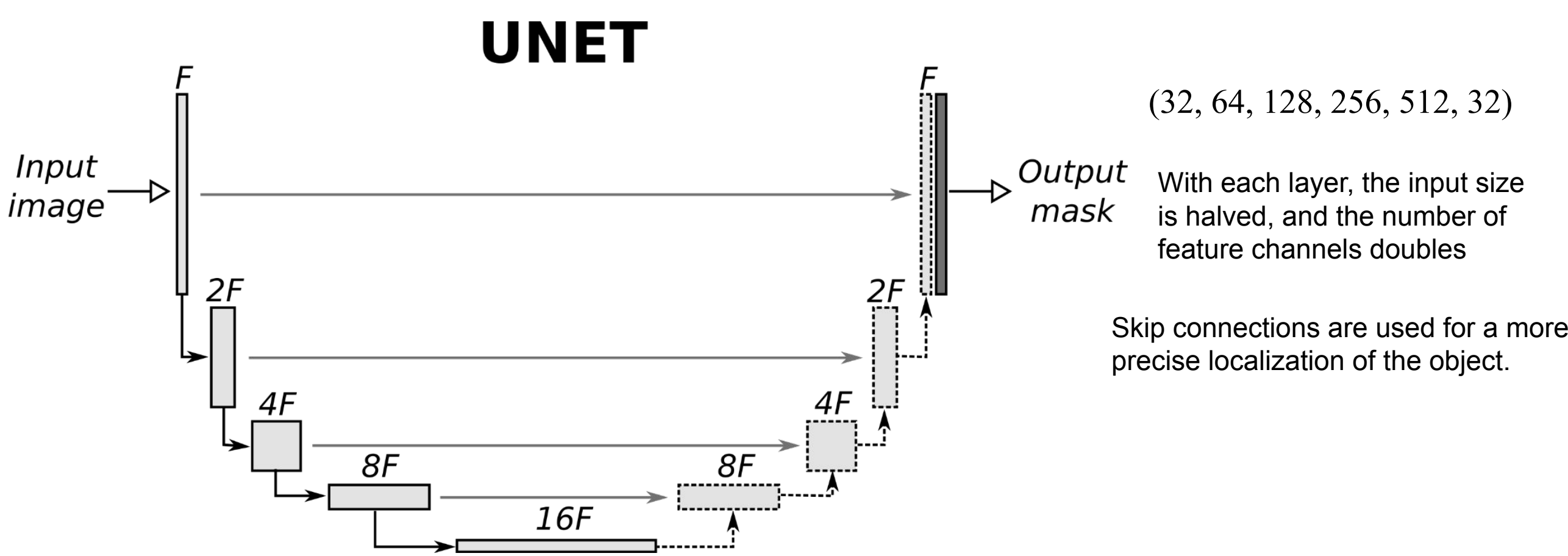
	U-Net	CNN
No FFT	 Average IoU for well 1 = 63.38%	 Average IoU for well 1 = 42.05%
FFT	 Average IoU for well 1 = 63.51%	 Average IoU for well 1 = 53.09%

Models

Convolutional Neural Network (CNN)



U-Net



Encoder Block:

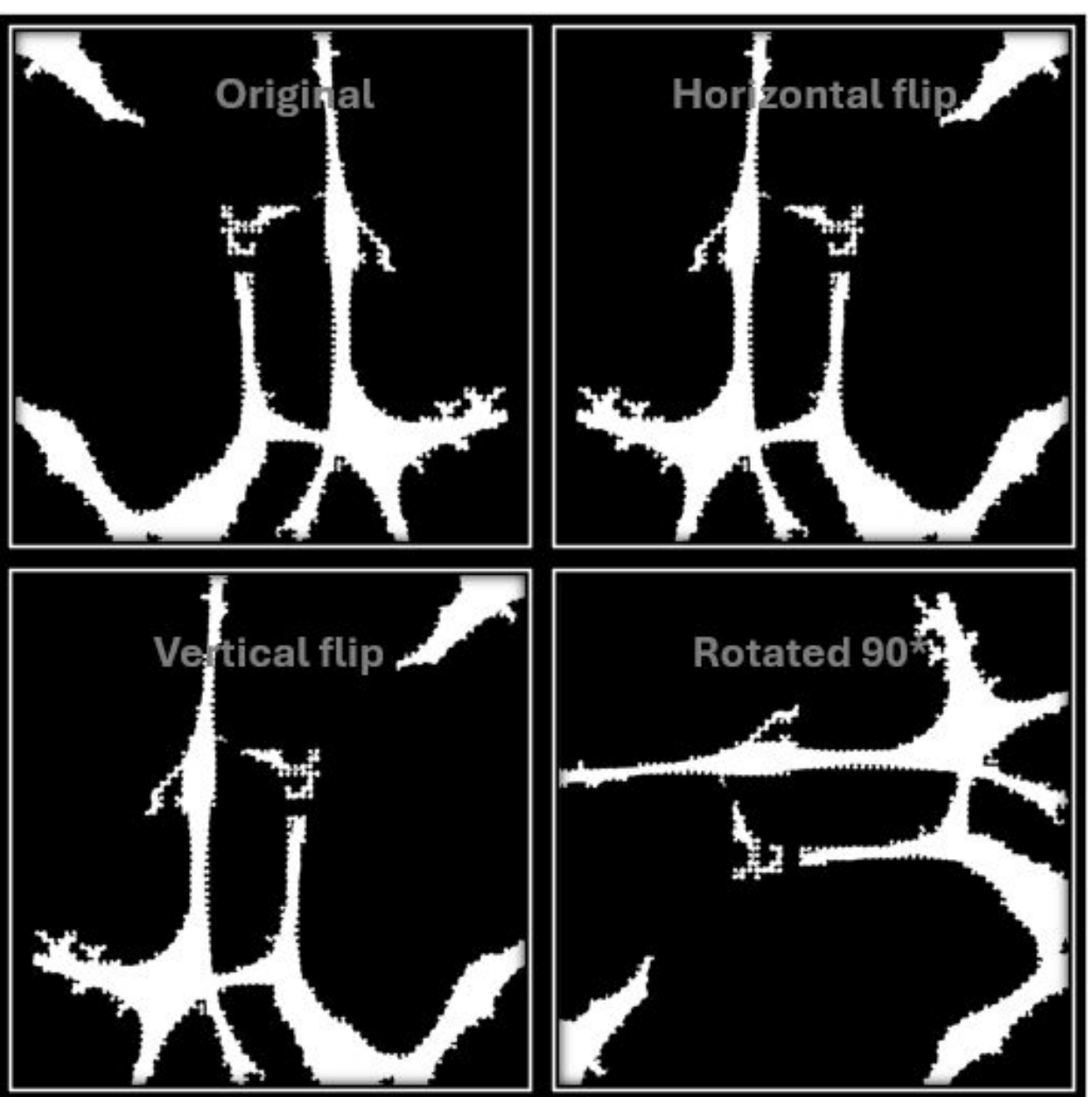
- **Block 1:** 32 filters, downsample by 2x (512x512)
- **Block 2:** 64 filters, downsample by 2x (256x256)
- **Block 3:** 128 filters, downsample by 2x (128x128)

Decoder Block:

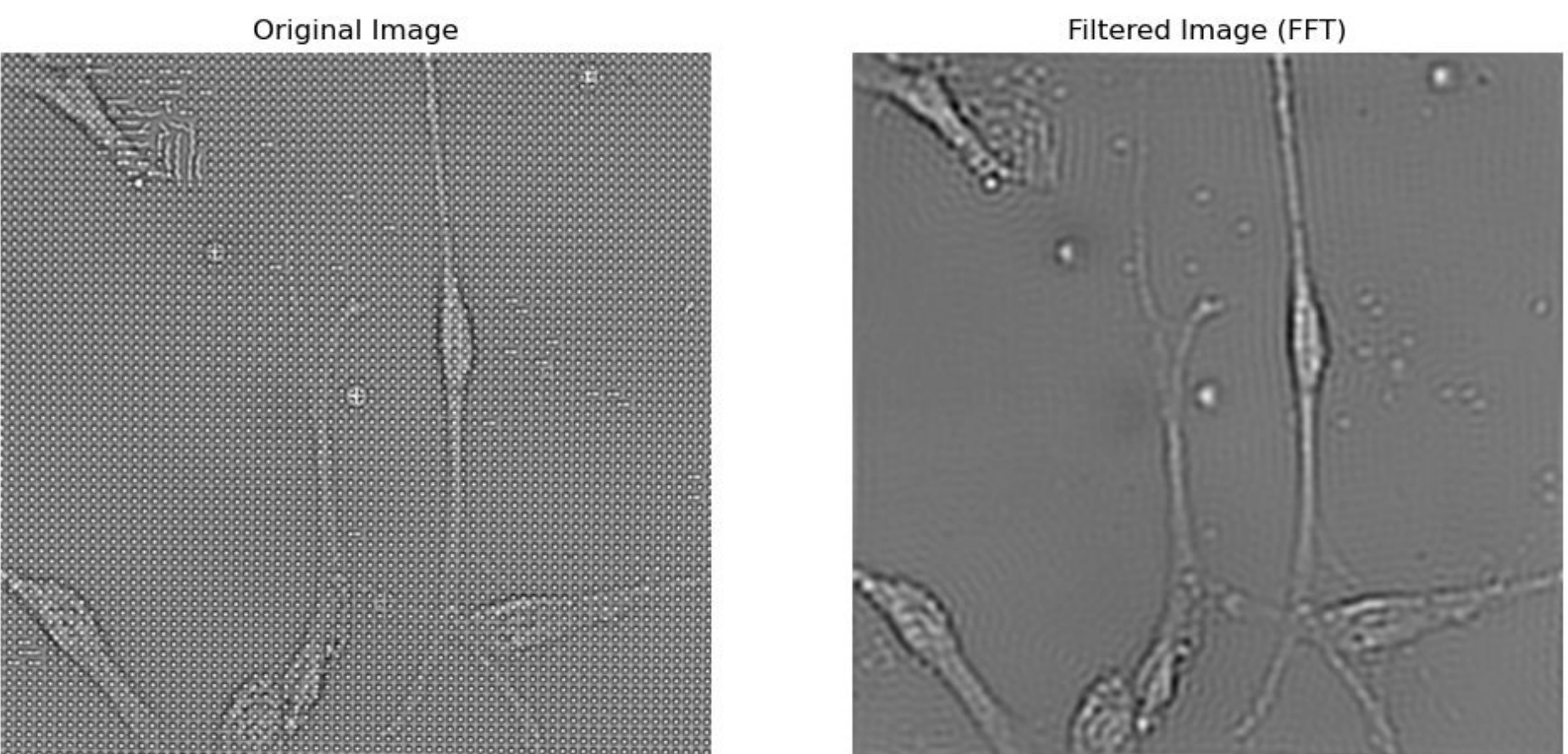
- **Layer 1:** Transpose convolution, 64 filters, upsample by 2x (256x256)
- **Layer 2:** Transpose convolution, 32 filters, upsample by 2x (512x512)
- **Layer 3:** Transpose convolution, 1 filter, upsample by 2x (1024x1024)

Pre-processing

○ augmentation



FFT Filtering



Discussions

Model Comparison:

- U-Net outperformed the CNN, achieving an **IoU of 63.51%** compared to the CNN's best of **53.09%**.
- U-Net's **skip connections** help retain spatial information, making it better for segmentation tasks.

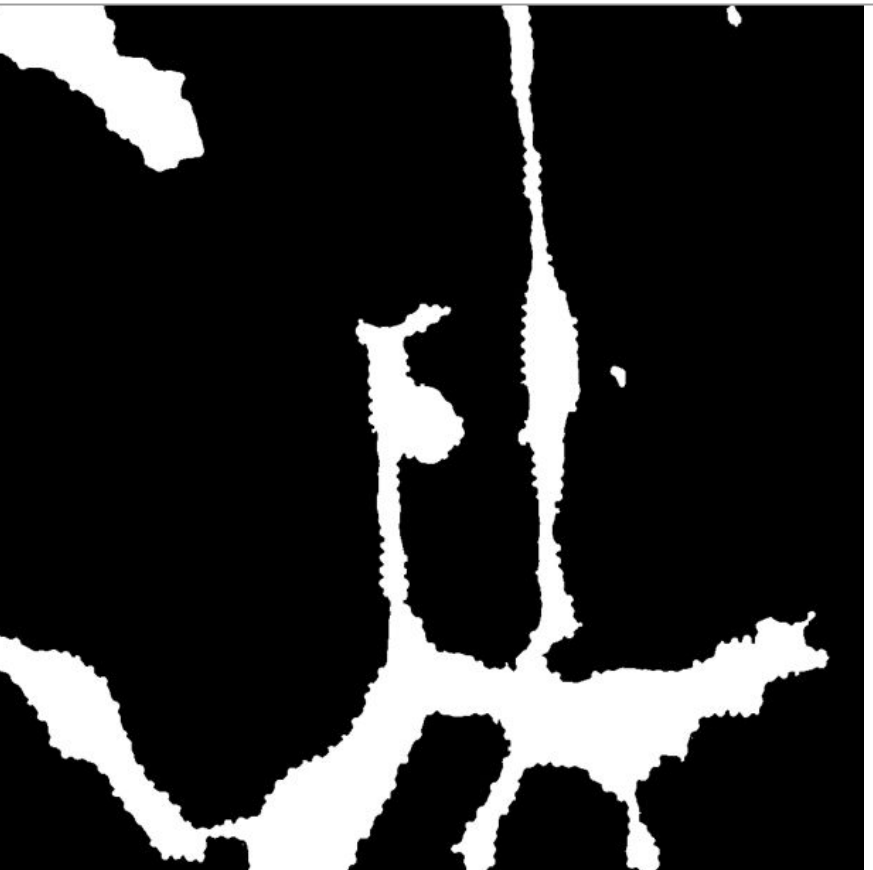
Noise Reduction with FFT Filtering:

- **FFT filtering** improved segmentation by reducing high-frequency noise from the brightfield images.
- CNN performance increased from **42.05% (BCE)** to **53.09% (BCE + FFT)**.
- U-Net showed a smaller improvement, increasing from **63.38% to 63.51%**, but masks had smoother cell boundaries.

Efficiency vs. Accuracy Trade-off:

- Using all **11 focal planes** yielded the best accuracy but required long training times (5498.43 seconds).
- Reducing to a **single focal plane** (6th layer) decreased IoU to **59.14%** but reduced runtime significantly (1539.93 seconds)

1 input image for U-Net



Average IoU for well 1 = 59.14%

